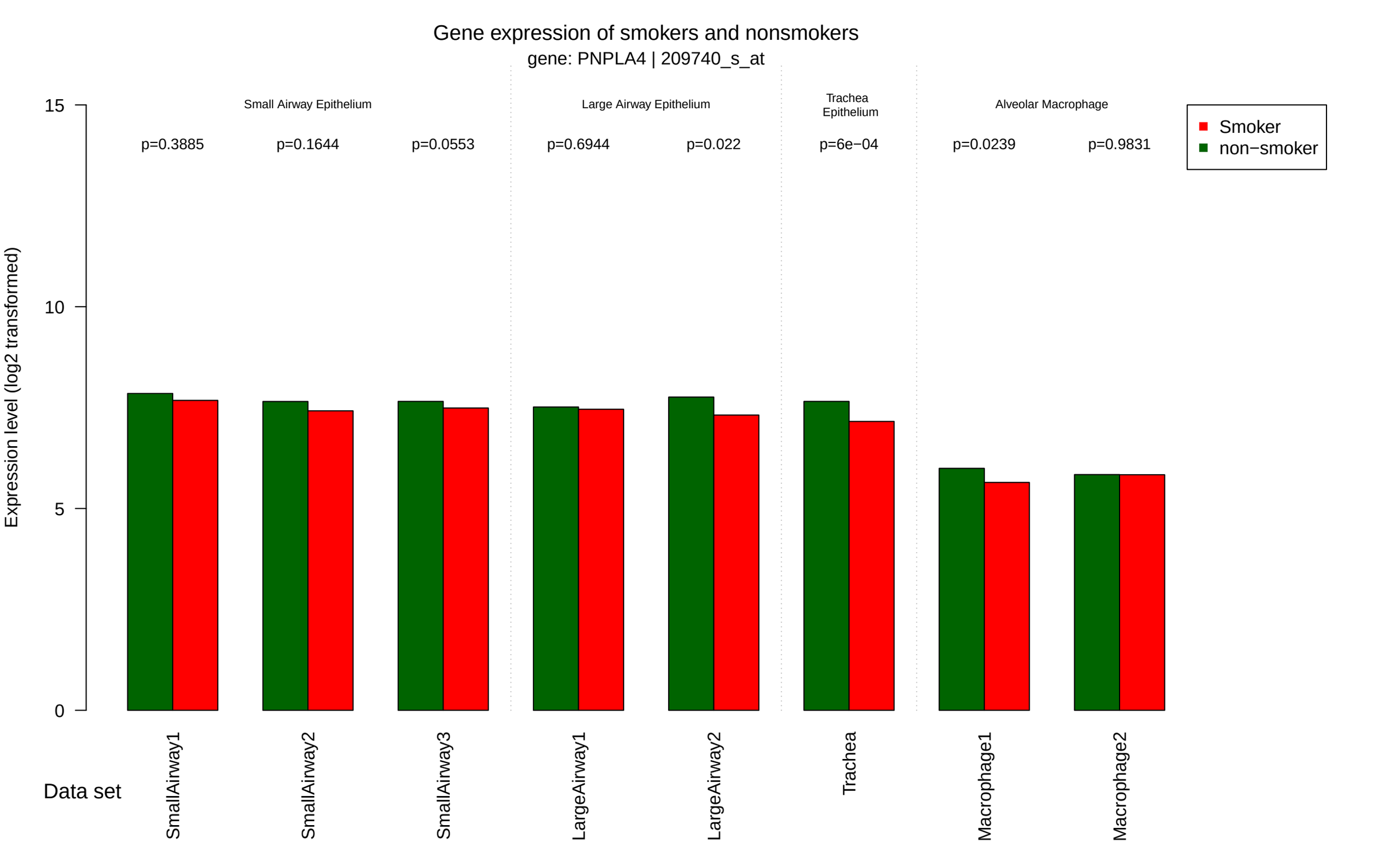
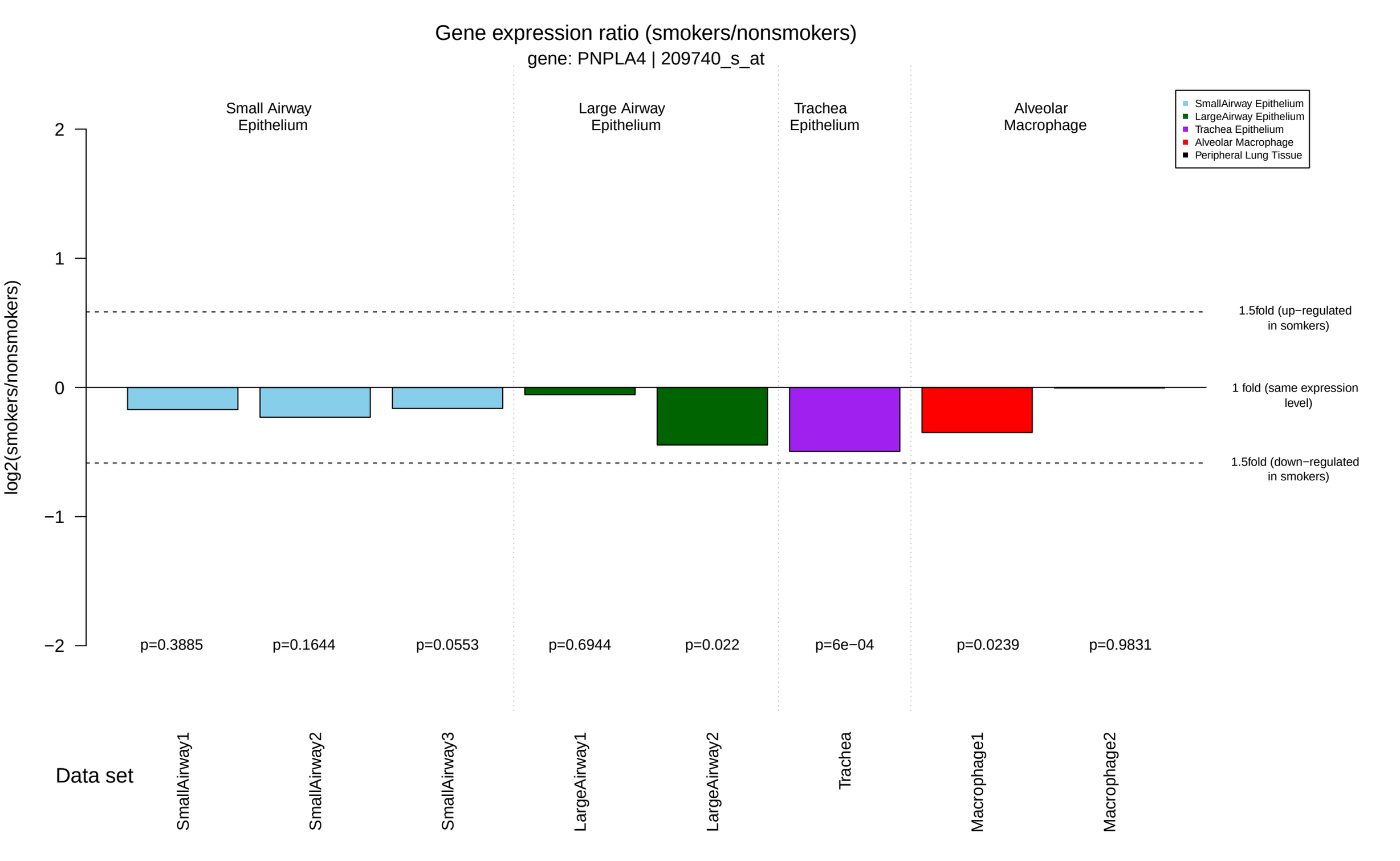
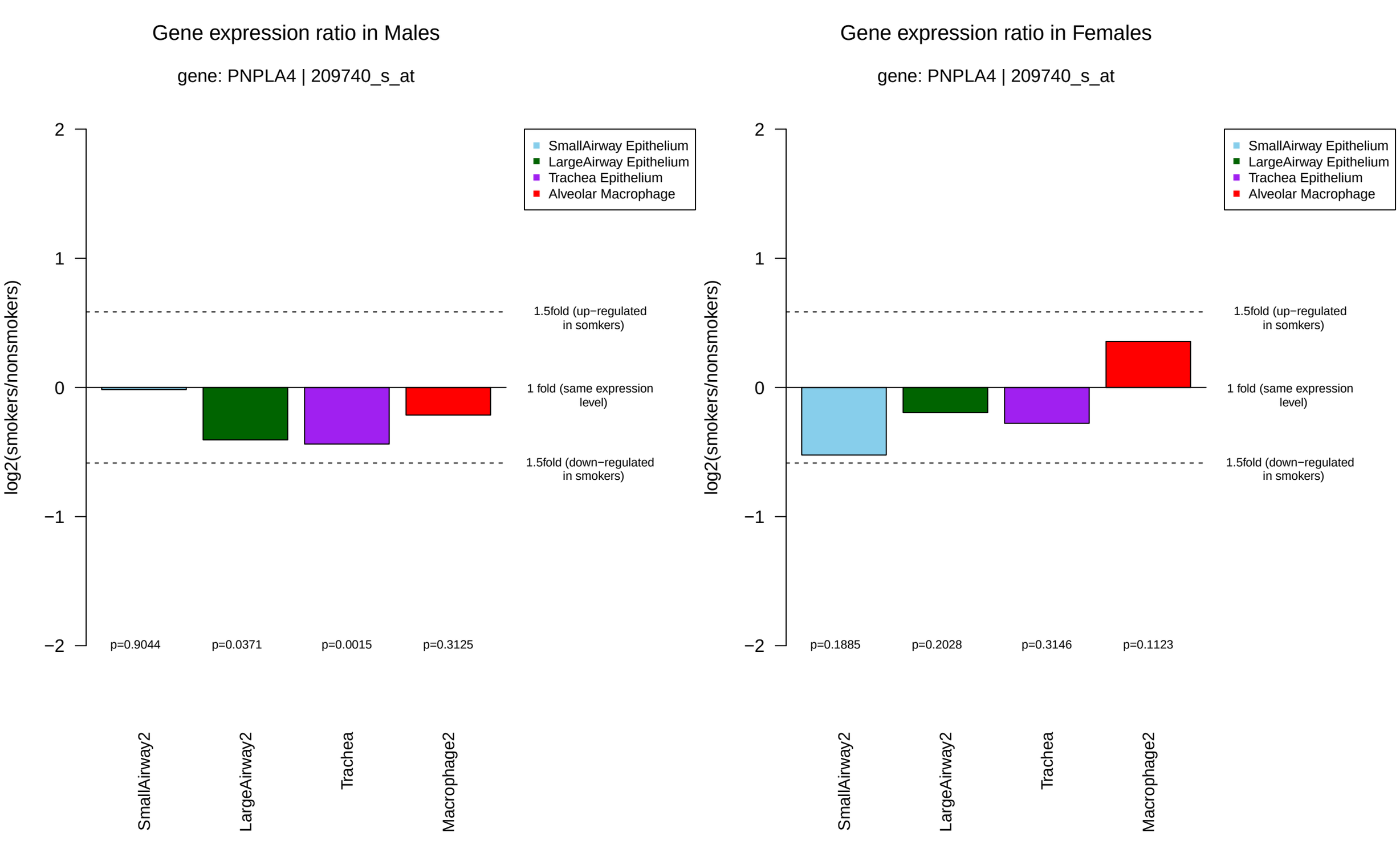
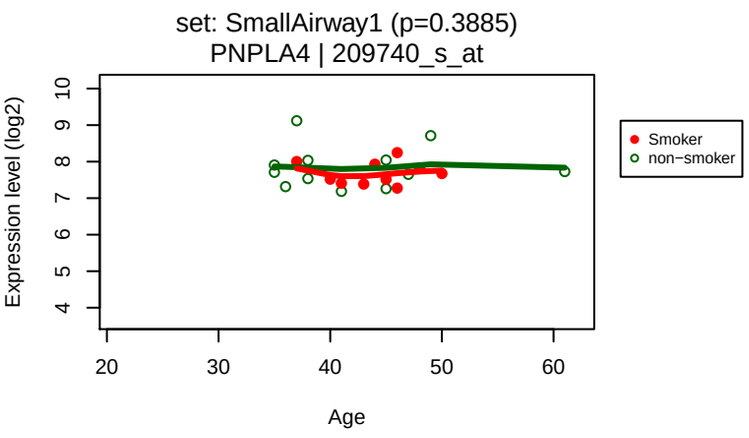
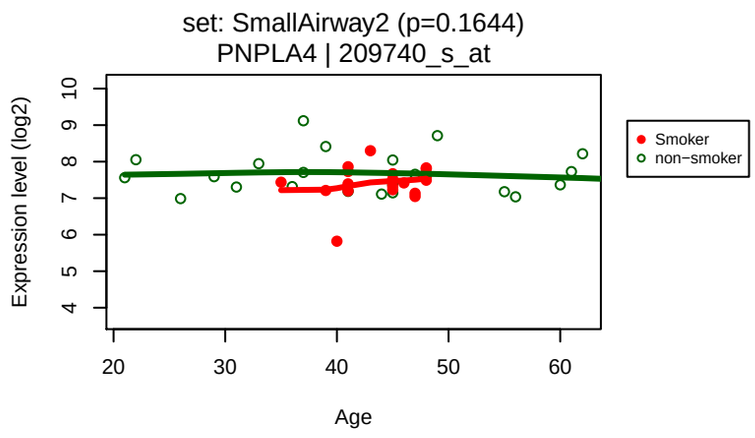
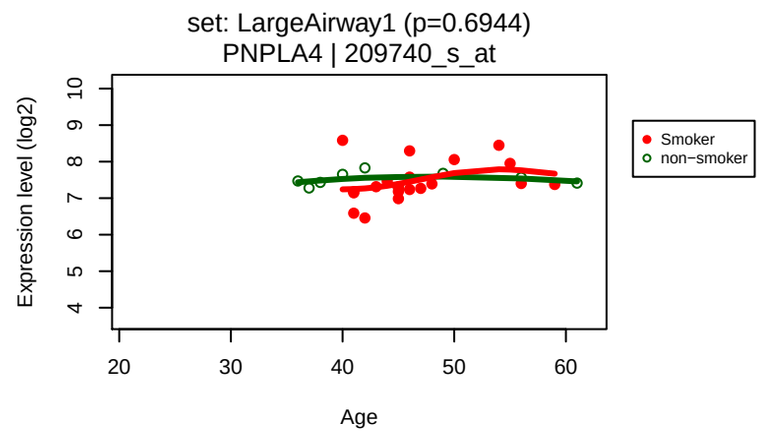

Gene expression of smokers and nonsmokers
gene: PNPLA4 | 209740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3885
p=0.1644
p=0.0553
p=0.6944
p=0.022
p=6e−04
p=0.0239
p=0.9831
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

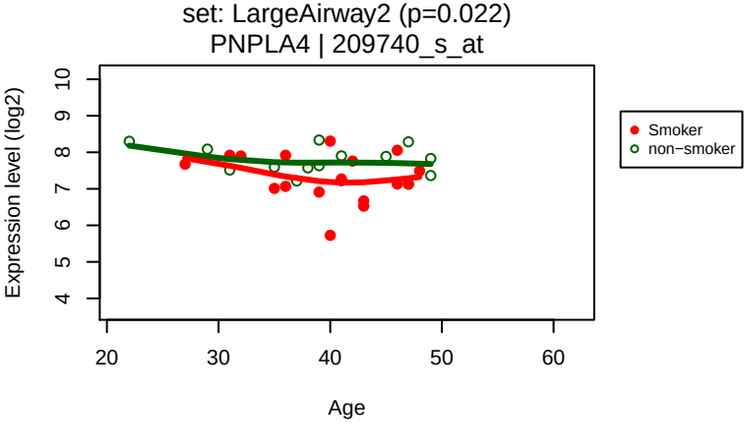

Gene expression ratio (smokers/nonsmokers)
gene: PNPLA4 | 209740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3885
p=0.1644
p=0.0553
p=0.6944
p=0.022
p=6e−04
p=0.0239
p=0.9831
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

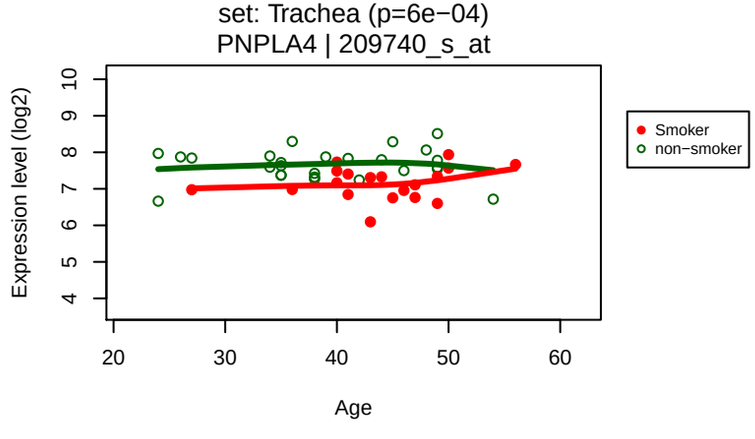

Gene expression ratio in Males
gene: PNPLA4 | 209740_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9044
p=0.0371
p=0.0015
p=0.3125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PNPLA4 | 209740_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1885
p=0.2028
p=0.3146
p=0.1123
SmallAirway2
LargeAirway2
Trachea
Macrophage2

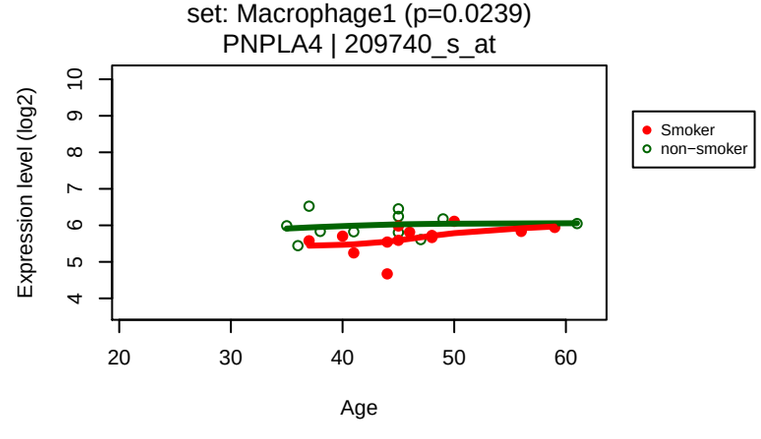

set: SmallAirway1 (p=0.3885)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker

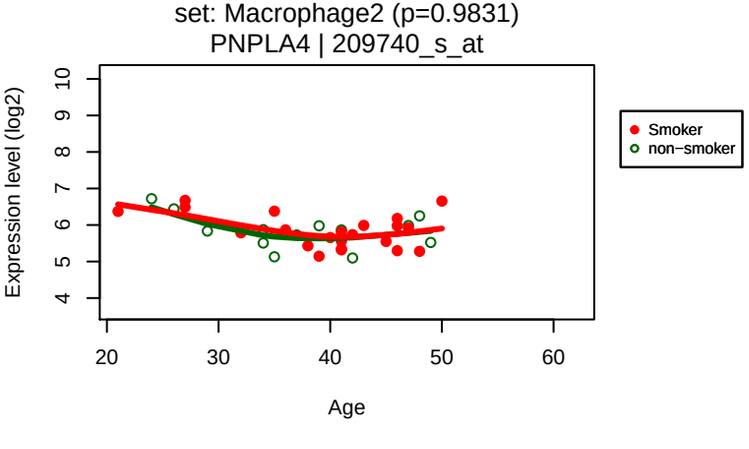

set: SmallAirway2 (p=0.1644)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker

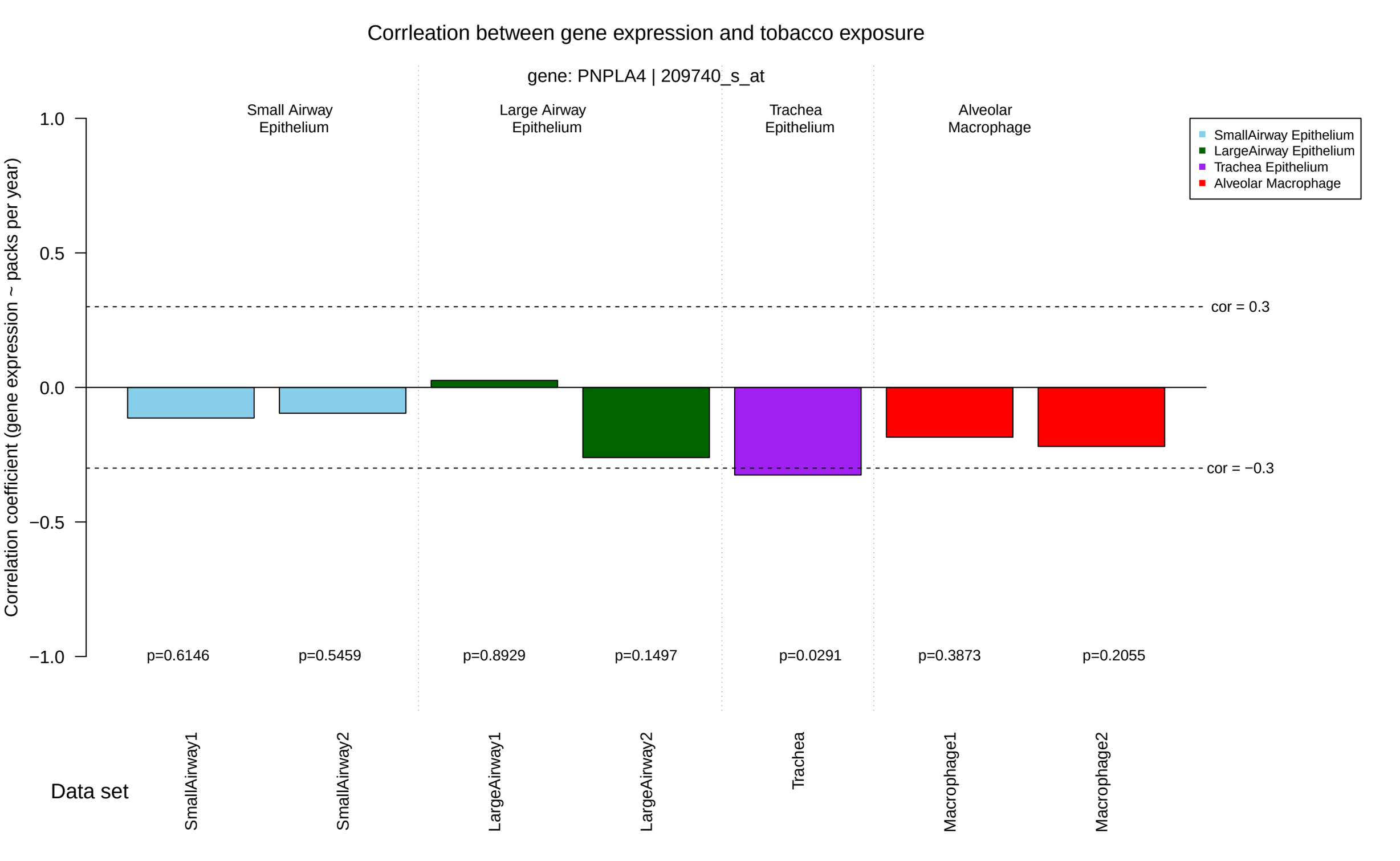

set: LargeAirway1 (p=0.6944)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.022)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=6e−04)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0239)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9831)
PNPLA4 | 209740_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PNPLA4 | 209740_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6146
p=0.5459
p=0.8929
p=0.1497
p=0.0291
p=0.3873
p=0.2055
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2